

Automatic Deburring

Below are **datasheet-style summaries** for the requested automatic deburring machines. Specs can vary by exact configuration, head layout, voltage, region, and options, so use these as a technical comparison starting point.

1. Timesavers 22 Series

Machine type: Automatic dry deburring / finishing / edge rounding machine

Best for: Laser-cut, punched, plasma/flame-cut sheet metal parts; general deburring, finishing, and heavy slag removal depending on head configuration.

Item	Datasheet
Manufacturer	Timesavers
Series	22 Series
Process type	Dry process
Main applications	Deburring, edge rounding, finishing, heavy slag removal
Working width	900 / 1050 / 1280 mm
Table opening	0–125 mm
Transport speed	2.5–14 m/min
Number of heads	1–3
Head types	Roller, brush, multi-disc, hammerhead
Abrasive belt length	1900 / 2620 mm
Contact roller diameter	150 / 200 mm
Power supply	400 V / 50 Hz
Certification	CE
Approx. weight	1150–4000 kg

[Catalog / product page](#)[Official Timesavers product page / brochure download page](#)

How it works: the part travels on a conveyor through one or more processing heads. A wide abrasive belt removes burrs and surface irregularities. Brush or multi-disc heads round the edges. For plasma/flame-cut parts with heavier slag, the Hammerhead option is used to knock off slag before finishing. Timesavers describes it as suitable for dry finishing and deburring, with three widths and up to three heads for multiple applications.

2. Timesavers 31 Series

Machine type: Automatic wet deburring and finishing machine

Best for: Heat-sensitive metal parts, sheet metal, square pipes, rectangular tube, and flat bar finishing.

Item	Datasheet
Manufacturer	Timesavers
Series	31 Series
Process type	Wet process
Main applications	Deburring and finishing
Working width	450 / 900 mm
Table opening	0–150 mm for 900 mm model; 0–200 mm for 450 mm model
Transport speed	2.5–14 m/min
Number of heads	1–3
Head types	Roller, brush
Brush diameter	150 mm
Contact roller diameter	150 mm
Abrasive belt length	1525 / 1900 mm
Main drive motor	7.5 / 11 / 15 kW

Table height	980 mm
Catalog / product page	Official Timesavers 31 Series page

How it works: the 31 Series uses a grinding belt and/or brush head to remove burrs and create a consistent surface finish. Because it is a wet-working system, coolant helps control heat, extend abrasive belt life, and protect heat-sensitive parts. The integrated coolant filter recirculates filtered coolant, and an integrated dryer dries parts at the exit.

3. Timesavers 32 RB Series

Machine type: Automatic rotary brush deburring / edge rounding machine

Best for: Broad sheet-metal deburring, edge rounding, laser oxide removal, finishing, and some heavy slag removal.

Item	Datasheet
Manufacturer	Timesavers
Series	32 RB Series
Process type	Dry rotary brush process
Main applications	Deburring, edge rounding, finishing, laser oxide removal, heavy slag removal
Working width	1100 mm
Table opening	0–100 mm
Transport speed	0.2–8 m/min
Number of heads	1–3
Head types	Hammerhead, wide belt, rotary brush
Number of brushes	4
Brush diameter	350 mm
Contact roller diameter	150 mm
HMI	Siemens

Certification	CE
Approx. weight	2500–3500 kg
Catalog / product page	Official Timesavers 32 RB Series page

How it works: the 32 RB Series combines abrasive belt and rotary brush technology. Burrs can be removed by the belt or hammerhead system, then rotary brushes round the edges from multiple directions. This is useful because cut sheet-metal parts often have burrs on complex contours and internal cutouts. Timesavers says deburring, edge rounding, finishing, and laser oxide removal can be achieved in one pass depending on configuration.

4. Timesavers 41 Series / 41 Series PUMA

Machine type: Heavy-duty automatic wet deburring and finishing machine

Best for: High-volume industrial production, 24/7 operation, thin/oily/greasy or potentially flammable metal parts.

Item	Datasheet
Manufacturer	Timesavers
Series	41 Series / PUMA
Process type	Wet process
Main applications	Deburring, finishing, precision grinding depending on configuration
Production class	Heavy-duty, 3-shift / 24/7 operation
Number of heads	Up to 4
Head types	Multiple head types; wide abrasive belt, disc, brush options depending on configuration
Typical working widths	900 / 1100 / 1350 mm in Timesavers Europe literature; 37 in / 52 in mentioned for PUMA in U.S. distributor material
Suitable materials	Thin, oiled, greasy, and/or highly flammable metal parts and materials

How it works: the 41 Series is the heavy-production version. It is a wet grinder/deburring system, so coolant helps control heat and dust while improving abrasive life. The machine can be configured with multiple heads for aggressive deburring, finishing, or precision grinding. Timesavers states the 41 Series PUMA is designed for 3-shift and 24/7 operation and can be built with up to four heads.

5. LISSMAC SMD 123 RE

Machine type: Automatic dry deburring, edge rounding, and surface grinding machine

Best for: Entry-level automatic processing of laser-cut, punched, and plasma-cut parts.

Item	Datasheet
Manufacturer	LISSMAC
Model	SMD 123 RE
Process type	Dry grinding
Main applications	Deburring, uniform edge rounding, surface finishing
Working width max.	950 mm
Material thickness	1–50 mm
Feed speed	0.5–8 m/min
Load capacity	300 kg/rm
Voltage, standard	400 V / 50 Hz
Voltage, USA	480 V / 60 Hz
Total power, standard	19.9 kW
Total power, USA	20.3 kW
Total current, standard	42.3 A
Total current, USA	42.7 A

Protection class	IP42
Material thickness adjustment	Electric
Tool adjustment	Manual
Weight	1750 kg
Dimensions	1600 × 1800 × 1950 mm
Catalog / datasheet	Official LISSMAC SMD 123 RE page with downloadable EN datasheet and manual

How it works: the SMD 123 RE uses a grinding belt and two rotary wheels. The belt removes burrs and performs surface grinding, while the rotary wheels create consistent edge rounding and uniform finishing around outer and inner contours. LISSMAC describes it as a cost-effective basic machine for surface finish and sheet-metal deburring, suitable for steel, stainless steel, and aluminum.

6. NS Máquinas DM1100 Z

Machine type: Automatic single-sided deburring and surface finishing machine

Best for: Light burr removal and surface finishing on laser-cut, punched, or sheared metal sheets up to 1100 mm wide.

Item	Datasheet
Manufacturer	NS Máquinas
Model	DM1100 Z
Process type	Dry abrasive belt deburring
Main applications	Deburring and surface finishing
Working capacity	1100 × 120 mm
Minimum part size	50 × 50 mm
Feeding speed	1.5–10 m/min
Abrasive belt speed	8–22 m/s
Abrasive belt size	1100 × 1850 mm

Feeding motor	0.55 kW
Abrasive belt motor	11 kW
Consumption	22 A
Dimensions	1200 × 1700 × 1990 mm
Weight	1090 kg
Contact roller	Ø186 mm, 55 SH
Catalog / datasheet	Official NS Máquinas DM1100 Z PDF technical file

How it works: the DM1100 Z uses one wide abrasive belt station with a large-diameter contact roller. The part passes under the belt, and the belt removes burrs and can also create a surface finish. The machine is simple compared with multi-head edge-rounding systems, so it is good for basic deburring and finishing. NS says it can process parts as small as 50 × 50 mm and sheets up to 1100 mm wide.

7. NS Máquinas DM1600 Z

Machine type: Automatic single-sided deburring and surface finishing machine

Best for: Wider sheet-metal parts up to 1600 mm, light burr removal, and surface finishing.

Item	Datasheet
Manufacturer	NS Máquinas
Model	DM1600 Z
Process type	Dry abrasive belt deburring
Main applications	Deburring and surface finishing
Working capacity	1600 × 120 mm
Minimum part size	50 × 50 mm
Feeding speed	1.5–10 m/min
Abrasive belt speed	8–22 m/s

Abrasive belt size	1600 × 1850 mm
Feeding motor	0.37 kW
Abrasive belt motor	15 kW
Consumption	25 A
Dimensions	1420 × 2190 × 1990 mm
Weight	1780 kg
Contact roller	Ø186 mm, 55 SH
Catalog / datasheet	Official NS Máquinas DM1600 Z PDF technical file

How it works: the DM1600 Z is basically the wider-capacity version of the DM1100 Z. It uses an abrasive belt station with a large contact roller to remove burrs and finish the surface. It is intended for stainless steel, steel, and aluminum parts cut by laser, punching, or shearing, and can be integrated with NS Máquinas' DM1600 C edge-rounding machine for a more complete deburring + edge-rounding line.

Quick selection guide

Machine	Best choice when you need
Timesavers 22 Series	Economical automatic deburring, finishing, edge rounding, and possible slag removal
Timesavers 31 Series	Wet deburring/finishing for heat-sensitive parts and consistent surface finish
Timesavers 32 RB Series	Stronger all-around edge rounding, laser oxide removal, and rotary brush finishing
Timesavers 41 Series	Heavy-duty 24/7 wet deburring and finishing production
LISSMAC SMD 123 RE	Compact entry-level automatic deburring + edge rounding + surface finishing

NS DM1100 Z	Simple 1100 mm wide abrasive-belt deburring and finishing
NS DM1600 Z	Same concept as DM1100 Z, but for wider 1600 mm parts

give me the picture for each machine

Searching for NS images

give me the download link for each machines catalog pdf

I'll switch from pictures to catalog PDFs and check the official manufacturer pages first. For gated brochures, I'll give the closest official brochure/download page and mark it clearly.

Thought for 30s

Here are the **catalog / datasheet PDF download links** I found.

Machine	PDF download link	Note
Timesavers 22 Series	Timesavers Sheet Metal Processing Machines brochure PDF	Official Timesavers PDF brochure; includes Timesavers sheet-metal machines including 22/31/32 series.
Timesavers 31 Series	Timesavers Sheet Metal Processing Machines brochure PDF	Official Timesavers PDF brochure. It includes the 31 Series wet deburring/finishing machine.
Timesavers 32 RB / 32 Series	Timesavers Sheet Metal Processing Machines brochure PDF	Official Timesavers PDF brochure. The 32 RB product page also exists, but the brochure download is usually behind a form.
Timesavers 41 Series / PUMA	Timesavers 41 Series PUMA official product page	I found the official product page, but not a direct official PDF catalog from Timesavers. A third-party-hosted Timesavers metal-family

		brochure mentions the 41/PUMA, but it is not hosted on Timesavers' own site.
LISSMAC SMD 123 RE	Official SMD 123 RE datasheet PDF	Official LISSMAC 2-page datasheet PDF.
LISSMAC SMD 123 RE full catalog	Official LISSMAC Metal Processing catalog PDF	Full LISSMAC metal-processing catalog; includes the SMD 1 Series / SMD 123 RE.
LISSMAC SMD 123 RE manual	Official SMD 123 RE operating manual PDF	Operating manual, not a sales catalog. Useful for installation/operation details.
NS Máquinas DM1100 Z	Official DM1100 Z technical file PDF	Official NS Máquinas PDF technical file.
NS Máquinas DM1600 Z	Official DM1600 Z technical file PDF	Official NS Máquinas PDF technical file.
NS Máquinas deburring catalog / leaflet	NS Máquinas Metal Deburring Machines leaflet PDF	Broader deburring-machine leaflet, includes DM-series context. It appears to be NS Máquinas material, but hosted through Squarespace/static CDN rather than the main  S Máquinas - Tube Polishing, Deburring and Edge rounding machines - Tube Polishing, Deburring and Edge rounding machines Home domain.

Important note: For **Timesavers 41 Series / PUMA**, I could confirm the official product page, but I did **not** find a direct official PDF download link hosted by Timesavers.

give me the recent price for each machine

I'll verify current public prices/used-market listings because new industrial machine prices are often "request quote." I'll separate **confirmed public listing prices** from **estimated new-price**

ranges so it doesn't look more certain than it is.

Thought for 33s

Below is the **recent public-price status** for each machine. The main point: **most new automatic deburring machines are not sold with fixed online prices**. They are usually **RFQ / request quote**, because price changes by working width, number of heads, wet/dry system, vacuum/magnetic table, extraction, voltage, installation, training, and freight.

Recent price table

Machine	Recent public price found	New machine price status	Practical budget estimate
Timesavers 22 Series	No fixed public price found. Recent ex-display 2023 listings show "Price info / Call" .	Request quote	~\$80,000–\$180,000+ depending on width and heads
Timesavers 31 Series	Used Timesavers Grindingmaster 31-Serie-450-WWB listed at €16,000 + VAT .	Request quote	~\$120,000–\$250,000+ new, depending on wet system/head configuration
Timesavers 32 RB / 32 Series	2019 and 2025/ex-display listings found, but price shown as Request quote / Price info .	Request quote	~\$180,000–\$350,000+ new
Timesavers 41 Series / PUMA	Used 2020 PUMA 52MWT-DB-60 listed at \$79,950 , but it was marked out of stock .	Request quote	~\$250,000–\$500,000+ new
LISSMAC SMD 123 RE	Public listings found, but price shown as Request quote / Price info .	Request quote	~\$70,000–\$130,000+ new

NS Máquinas DM1100 Z	Dealer page shows “ Call for Pricing ”. UK listing showed £0.00 , meaning quote-only, not actual price.	Request quote / call	~\$45,000–\$90,000+ new
NS Máquinas DM1600 Z	UK listing showed £0.00 , meaning quote-only, not actual price.	Request quote	~\$65,000–\$130,000+ new

Source notes

For **Timesavers**, public marketplaces show many listings but most exact models are **request quote**. Machinesseeker shows the 22 Series and 32 Series as “Price info / Call,” while one 31-Series listing shows **€16,000 + VAT**.

For **Timesavers 41 Series / PUMA**, I found a 2020 PUMA 52MWT-DB-60 used listing at **\$79,950**, but it is marked **out of stock**, so treat it as a historical market reference, not an available current offer.

For **LISSMAC SMD 123 RE**, current listings are mostly **request quote**. The official LISSMAC page gives technical specs, while marketplace listings show the model but not a public fixed price.

For **NS Máquinas DM1100 Z / DM1600 Z**, dealer pages do not publish real prices. US Tool Depot shows “**Call for Pricing**” for DM1100 Z, and Sparx Machine Tools lists DM1100 Z and DM1600 Z at **£0.00**, which is a quote placeholder, not a selling price.

Best buying estimate: if you need a real quotation, expect the **NS DM1100 Z / LISSMAC SMD 123 RE** to be the lower-cost automatic options, while **Timesavers 32 RB and 41/PUMA** are much more expensive production machines.